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Hyperfine structure and g-factor measurements on Ba^+ and Eu^+ isotopes

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Paul and Penning ion traps have been used to determine hyperfine structure coupling constants as well as nuclear g-factors of Ba^+ ions with great precision. Attempts are under way to measure both quantities for a long chain of Eu isotopes in order to determine differential hyperfine anomalies for a better understanding of the Bohr–Weisskopf effect.

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1. Introduction

g-factors and hyperfine splittings (hfs) of atomic energy levels have been a traditional field of experimental atomic physics since many decades. The results serve as basis for the construction of atomic wavefunctions. Numerous data on stable and unstable isotopes of the elements exist, and generally the agreement between experimental data and atomic structure calculation is typically on the level of 1 part in thousand. Apart from atomic structure implications g-factors and hfs may also serve as input data for nuclear physics models. The magnetic dipole part of the hyperfine splitting of atomic energy levels, characterised by the magnetic dipole interaction constant A, is for a point-like nucleus proportional to the nuclear magnetic moment μ_{I} or the nuclear g-factor g_{I} ($\mu_{\mathrm{I}} = g_{\mathrm{I}}\mu_{\mathrm{N}}$, μ_{N} being the nuclear magneton).

$$A = \frac{16\pi}{3}\mu_{\mathrm{B}}^2 g_{\mathrm{I}} \left|\Psi(0)\right|^2 F. \tag{1}$$

Here μ_{B} is the Bohr magneton, $|\Psi(0)|^2$ the electron density at the nucleus and F a relativistic correction.

For extended nuclei, however, the distribution of magnetisation over the nuclear volume has to be taken into account. It causes a modification of A, called the Bohr–Weisskopf effect [1], which is empirically described by

$$A = A_{\mathrm{P}}(1 + \varepsilon). \tag{2}$$

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Nuclear models attempt to calculate ε usually using shell model nuclear wavefunctions [2–4]. Different models, however, differ in many cases substantially among each other [3]. From the experimental side ε cannot be measured, since there is no point-like nucleus for comparison. Accessible to measurements is a quantity ${}^1\Delta^2$, the differential hyperfine anomaly, which is defined as

$$ {}^1\Delta^2 = \frac{A_1}{A_2}\frac{g_{\mathrm{I},2}}{g_{\mathrm{I},1}} - 1, \tag{3} $$

where the subscripts 1,2 refer to the two isotopes of an element. Since ε is a small number of the order of 10^{-2}–10^{-3}, ${}^1\Delta^2$ is to a good approximation the difference $\varepsilon_1 - \varepsilon_2$. Renewed interest in a better understanding of the Bohr–Weisskopf effect arose from problems in the comparison between experimental and theoretical data in hyperfine measurements on highly charged hydrogen-like ions such as ${}^{209}\mathrm{Bi}^{82+}$ [5] or ${}^{207}\mathrm{Pb}^{81+}$ [6]. This calls for new measurements as well as improved calculations. In order to make a reasonable comparison to nuclear models it is desirable to have systematic data from several isotopes of one element. Up to present there is only one such example, where A-factors and nuclear magnetic moments have been measured for a chain of isotopes; this is the mercury atom. The values of Δ obtained in these measurements are of moderate agreement to nuclear models [7]. There is, however, a surprisingly good agreement to an empirical rule, obtained by Moscowitz and Lombardi (ML) [8], who set the value of ε inverse proportional to the value of μ_{I} with an empirical constant factor C. No justification of this rule in terms of nuclear models has been given so far. In order to check whether this rule has a more general application, it is required to perform similar measurements as in mercury on a different element.

We are aiming at a determination of the differential hyperfine anomaly Δ in a chain of europium isotopes. Eu offers a long chain of 9 isotopes with nonzero nuclear spin which live longer than one week. They can be produced by nuclear reactions at the ISOLDE facility at CERN and investigated off-line at our laboratory at Mainz. From the nuclear physics point of view, europium ($Z = 63$) should be a suitable candidate for model calculation since it can be described in the frame of a single particle model due to the one-proton hole in the 5d nuclear subshell. Moreover, the measured nuclear magnetic moments show a drastic change by roughly a factor of 2 at the mass number 153. This change would consequently appear in values for the hyperfine anomaly calculated from the ML-rule. Core polarisation calculations as performed by Asaga et al. [7], however, do not show this change. Thus, a comparison to measured values could be a decisive test for the calculations.

2. Experimental principles

An experimental determination of the differential hyperfine anomaly requires independent measurements of hyperfine coupling constants A and nuclear magnetic moments. We use the ion storage technique for our investigations since its potential for very precise hfs and g-factor measurements has already been demonstrated [9].

A determination of Δ at the 1% level of accuracy, which seems reasonable considering the status of the theory, requires a measurement of A-factors and nuclear magnetic moments to about 10^{-5} when we assume that Δ is typically of the order of 10^{-2}–10^{-3}.

3. Hyperfine measurements

In recent years we have performed hyperfine measurements on different ions. We used a radiofrequency (Paul) trap of 2 cm inner diameter, driven by an electric field of 1 MHz frequency and up to 1.5 kV amplitude. In this trap we confined about 10^5 ions for a time of several hours at a typical background pressure of 10^{-5} mbar N_2 buffer gas. The gas served to damp the ions oscillation amplitude in the trap, thus reducing the Doppler width of the optical transition, as well as to quench metastable ionic states into which a laser excited ion may decay. A laser beam passes through the trap in midplane and a photomultiplier tube served to detect laser induced fluorescence photons perpendicular to the laser beam. For this purpose one trap endcap electrode was made from a transparent mesh. Figure 1(a) shows a sketch of the set-up. To detect microwave induced transitions between different ground state hyperfine levels we depleted one of these levels by selective laser excitation. Induced $\Delta F = \pm 1$ transitions were detected by increase of the fluorescence intensity. In the case of Eu^+ this is rather difficult, since the electronic ground state of this ion has a J quantum number of 4. Together with nuclear spin of 5 or $5/2$, depending on the isotope, this leads to many hyperfine sublevels which, moreover, show a Zeeman splitting in the weak ambient magnetic field in our laboratory. The high density of states makes selective excitation difficult. We tested, therefore, our technique on Ba^+ ions [10–12].

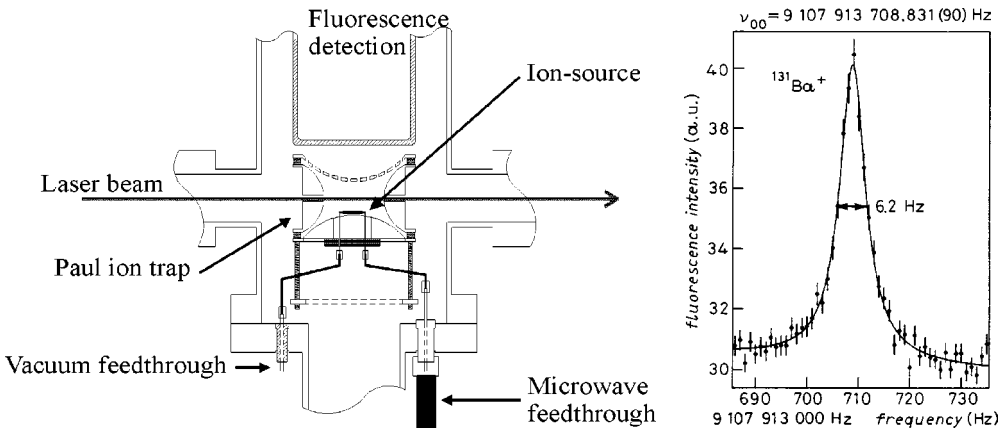


Figure 1. (a) Set-up of a Paul trap to determine hyperfine separations. (b) Microwave induced $\Delta F = 1$, $\Delta m_F = 0$ hyperfine transition in the ground state of $^{131}Ba^+$ ($\tau_{1/2} = 11$ d). The fractional uncertainty of the line center is $1 \cdot 10^{-10}$.

Table 1
Ground state magnetic dipole hyperfine coupling constants A for different Ba^+ isotopes, measured in a Paul ion trap.

Ba^+-isotope	Nuclear spin	Half life	A [Hz]
137	$3/2$	∞	4 018 870 833.85(18)
135	$3/2$	∞	3 591 670 117.45(29)
133	$1/2$	10.7 a	9 925 453 554.59(10)
131	$1/2$	11.8 d	9 107 913 698.97(50)

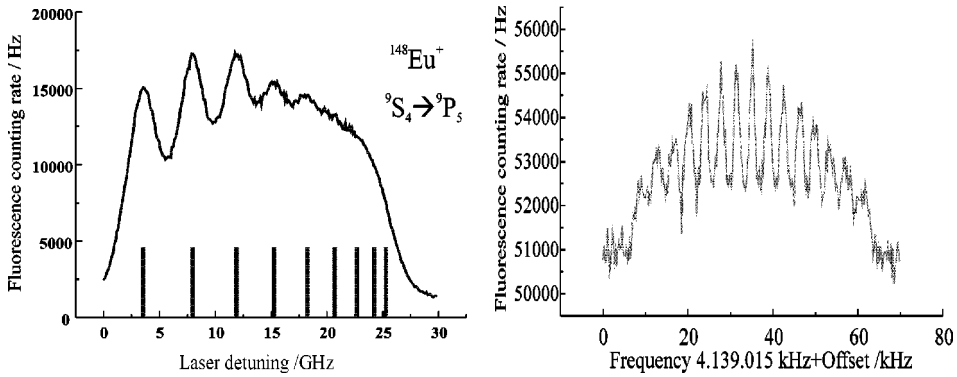


Figure 2. (a) Hyperfine structure of the $^9S_4$–$^9P_5$ resonance transition of $^{148}Eu^+$. (b) Microwave induced $F = 8 \rightarrow F = 7$, $\Delta m_F = 0$ transitions in the $^9S_4$-state of $^{148}Eu^+$. The line shows a Zeeman splitting in a residual field of 6 μT. Fractional uncertainty of transition frequency at zero magnetic field is $2 \cdot 10^{-9}$.

They exhibit a level scheme similar to alkali atoms, having an $S_{1/2}$ ground state. Thus, the Zeeman splitting can simply be handled by the well-known Breit–Rabi formula. The nuclear spins of the odd stable and long lived radioactive isotopes are $3/2$ or $1/2$. The state density is low and optical pumping is easy. In microwave induced hyperfine transitions we obtained linewidth of the order of a few Hz at transition frequencies of several GHz (figure 1(b)) from which A-factors were determined with a fractional uncertainty of the order of 10^{-10}.

Equally important was the fact that we were able to perform measurements with about the same accuracy on radioactive isotopes. The shortest lived isotope which we could investigate at our laboratory in Mainz was $^{131}Ba^+$, which has a half life time of 11 days and which was available in a quantity of only 10^{12} particles collected on a filament at ISOLDE/CERN. This demonstrated the applicability of the ion storage technique for very rare isotopes without loss of accuracy. Table 1 shows the results for different Ba^+ isotopes.

In the case of the Eu^+ isotopes we applied the same principle of laser-microwave double resonance [13,14]. Due to the complexity of the hyperfine structure (figure 2(a)), the large number of Zeeman-substates and configuration mixing of different energy levels we had to measure several ΔF transitions for each isotope in order to account

Table 2
Magnetic dipole hyperfine coupling constants A in the $^9S_4$ electronic ground state of Eu^+ isotopes, measured in a Paul ion trap.

Eu^+-isotope	Nuclear spin	Half life	A [Hz]
153	$5/2$	∞	648 565 993(9)
151	$5/2$	∞	1 540 297 394(13)
150	5	36 a	559 010 680(200)
149	$5/2$	93 d	1 585 450 570(258)
148	5	54 d	517 281 950(158)

for electric quadrupole interaction and higher order perturbation. Figure 2(b) gives an example of microwave induced $\Delta m_F = 0$ hyperfine transitions between $F = 8$ and $F = 7$ levels in the ground state of the unstable isotope $^{148}Eu^+$ (half life 54 d). The experimental uncertainty was about one order of magnitude lower than for the simpler Ba^+ isotopes. It was sufficient, however, to determine A-factors with an accuracy of about 10^{-7} for the two stable isotopes of mass 153 and 151 as well as the unstable isotopes of mass 150, 149 and 148 (table 2).

The accuracy exceeds by far what is required for a determination of the differential hyperfine anomaly in those isotopes as discussed above.

4. g-factor measurements

For the determination of g-factors the application of a magnetic field is required. The field strength should be as high as possible to make the Zeeman splitting large. This leads to high transition frequencies and consequently for a given linewidth to small fractional uncertainties. This suggests the use of static Penning ion traps with a superimposed strong magnetic field. In our experiments we used a field from a superconducting solenoid. The hyperbolic shaped trap of 2 cm inner diameter is placed in the horizontal room temperature bore of the magnet. A laser beam enters the apparatus off axis, is reflected into the trap through a hole in the ring electrode by a mirror and backreflected into itself by a second mirror at the opposite side of the ring. The fluorescence light is focussed through an endcap electrode onto a quartz fiber and guided to a photomultiplier tube about 1 m apart from the magnet. Figure 3(a) shows the set-up. As in the case of the hyperfine measurements we tested the performance of our apparatus by using Ba^+ ions because of their simpler level structure [15]. For the even isotope $^{138}Ba^+$ we induced $\Delta m_J = 1$ transitions after depletion of one of the ground state Zeeman levels by optical pumping. At a magnetic field strength of $B = 3$ T, corresponding to a transition frequency of 80 GHz, we obtained a linewidth of a few kHz. After calibration of the magnetic field by the cyclotron frequency of trapped electrons we determined the electronic g_J-factor to a precision of 10^{-8}. In order to measure the nuclear g_I-factor of an odd isotope, which is the quantity of interest for the hyperfine anomaly, we performed a triple resonance experiment: A broad band laser depletes the nuclear Zeeman manifold of the electronic $m_J = 1/2$ Zeeman level.

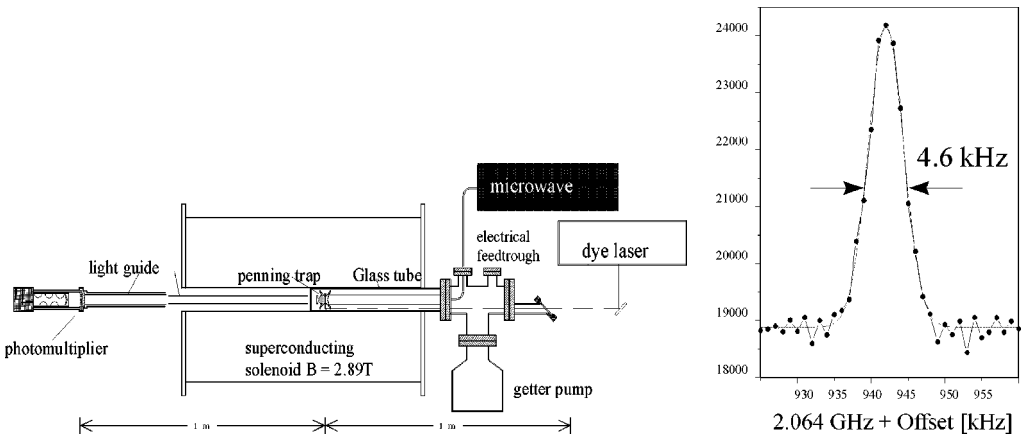


Figure 3. (a) Penning trap set-up for measurement of g-factors. (b) $\Delta m_J = 0$, $\Delta m_{\rm I} = 1$ nuclear Zeeman transition in $^{135}Ba^{+}$.

A $\Delta m_J = 1$, $\Delta m_{\rm I} = 0$ transition empties a single nuclear Zeeman substate of the $m_J = -1/2$ manifold. A third radiofrequency field corresponding to a $\Delta m_{\rm I} = 1$ or $\Delta m_{\rm I} = -1$ nuclear Zeeman transition is detected by the increase of fluorescence intensity at optical excitation (figure 3(b)). Using the Breit–Rabi formula and the hyperfine A-factor determined previously we obtained for the nuclear g-factor a value

$$g_{\rm I}\left(^{135}{\rm Ba}\right) = 0.623\,876(3).$$

This value improves a previously measured value, determined by Olschewski and Otten [16] using optical pumping of neutral Ba in a cell, by one order of magnitude. A further improvement seems possible. This experiment demonstrates the feasibility of direct nuclear g-factor measurement using Penning ion traps.

The case of Eu^{+} isotopes again is much more complicated because of the high angular momenta and complex level structure of the $^9S_4$ ground state. At present we have observed laser induced fluorescence from stored Eu^{+} ions and are trying to identify individual nuclear Zeeman substates in order to perform double resonance experiments on these ions (figure 4). A particular feature of this experiment is the use of buffer gas which serves to cool the ions by viscous drag and at the same time deexcite several long lived metastable states into which the laser excited states decay. To prevent ion loss from the trap as a consequence of collisions which tend to increase the magnetron radius of the ions we apply an additional radiofrequency field between adjacent segments of the ring electrode. The frequency of this field is set at the sum of the reduced cyclotron and the magnetron frequency. It couples the two eigenmotions of the ions in the trap. Buffer gas collisions reduce the amplitude of the reduced cyclotron motion. Due to the coupling the radius of the magnetron motion is also decreased and the ions are forced toward the trap centre. This technique, applied first time in mass spectrometry using Penning traps by Savard et al. [17] has been successfully applied

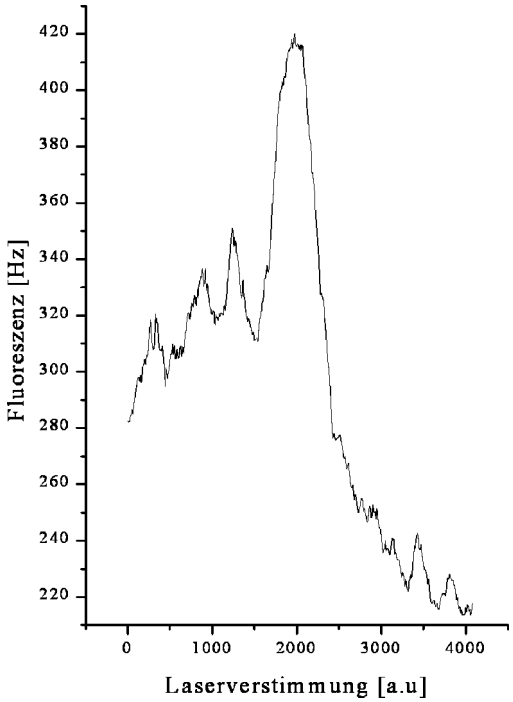


Figure 4. Laser-induced fluorescence from stored Eu^+ ions in a Penning trap with partially resolved Zeeman levels.

in our experiment on Ba^+ ions [18]. In the case of Eu^+ we succeeded to increase the storage time of the ions from about 15 min under ultra-high vacuum conditions to more than one week when we used 10^{-5} mbar of N_2 as buffer gas. Since the coupling frequency of the additional radio-frequency field depends on the mass of the stored ions we expect to separate the two stable isotopes of Eu^+ of mass 151 and 153, which are confined simultaneously in the trap with about equal amounts, by proper choice of background gas pressure and radio-frequency amplitude. Keeping one isotope in the trap while the other leaves it after a short time as the consequence of collisions would greatly simplify the optical spectrum, which at present is a convolution of hyperfine splittings and isotope shift. We expect to resolve the Zeeman components at a magnetic field strength of 1.5 Tesla which we choose in order to observe at least two electronic Zeeman components of the $^9S_4$–$^9P_5$ resonance line in one laser scan. The total span of the Zeeman components of this line in our magnetic field is 140 GHz which is too large for the mode-hop-free scan range of our frequency doubled Ti:Sa laser.

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